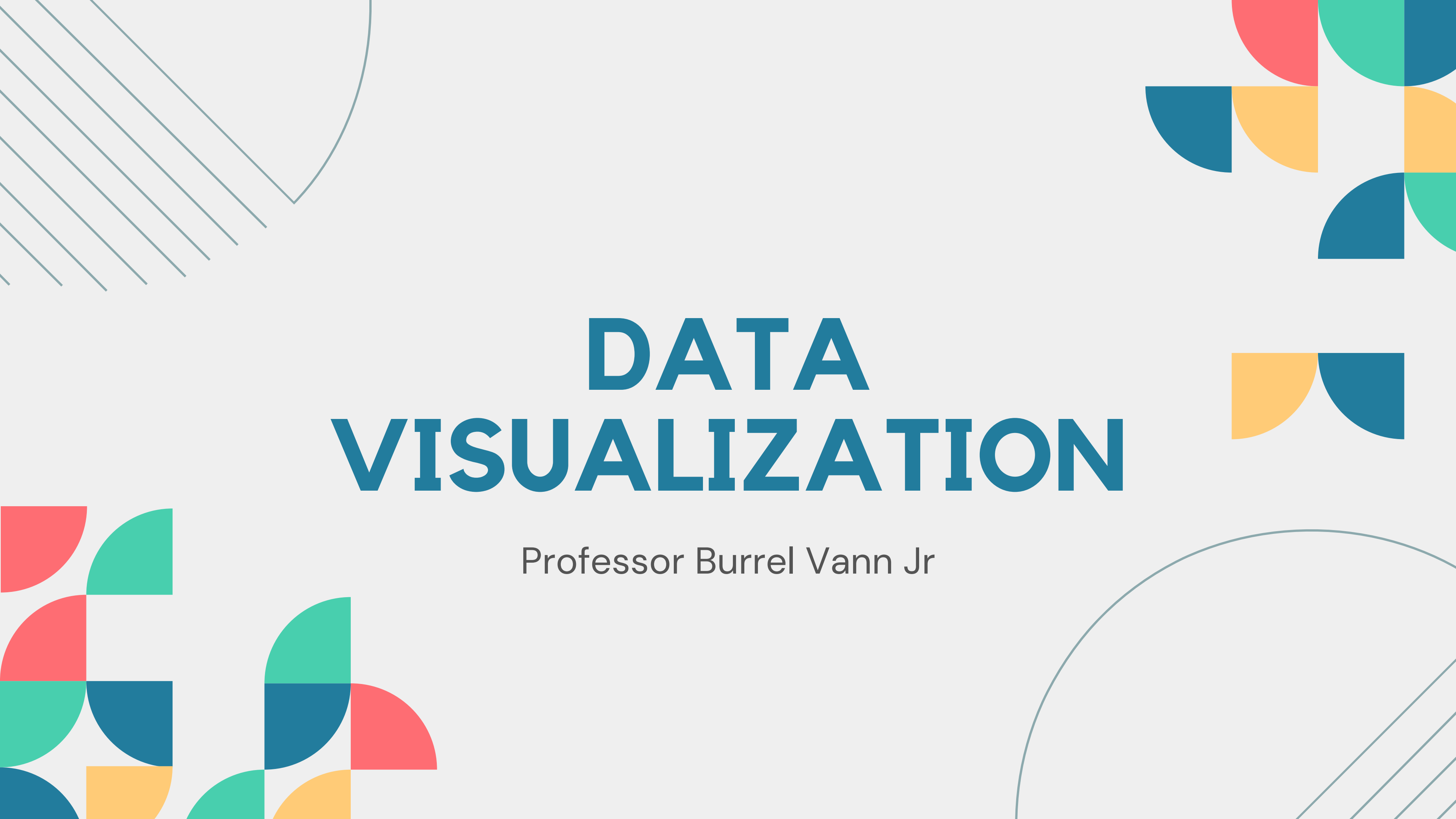




DATA VISUALIZATION

Professor Burrel Vann Jr

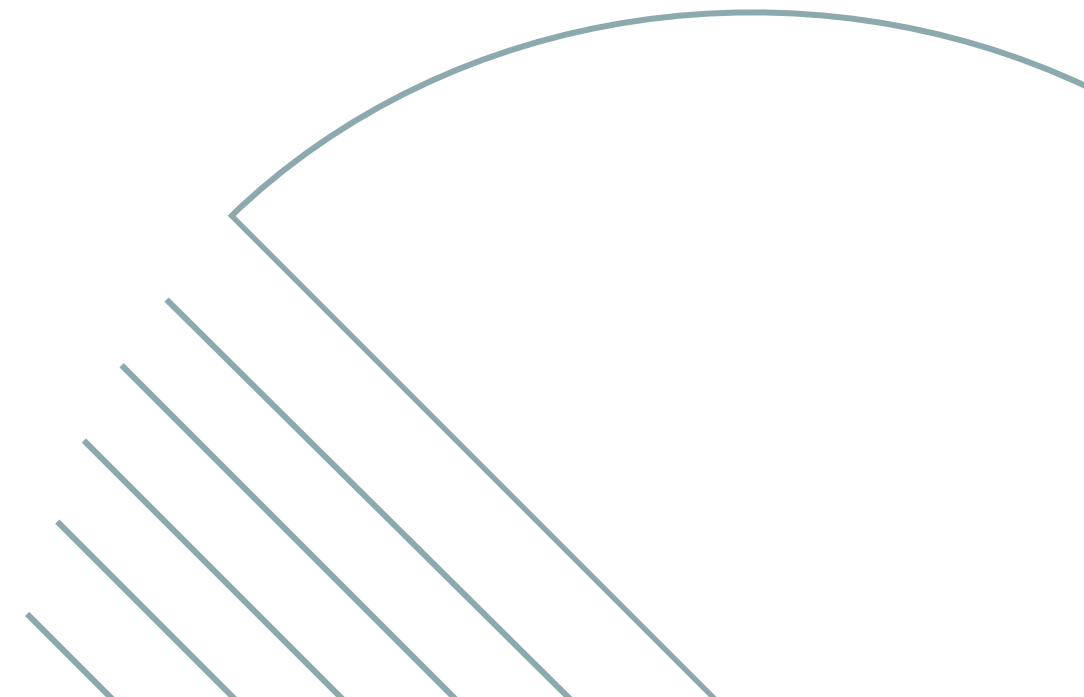
WHY VISUALIZE DATA?

- Data visualization simply refers to graphical displays (e.g. charts, graphs) of your data
- Visualizations can help people more easily see and understand patterns in the data (rather than looking at the whole data set itself)
- Government managers, public servants, law enforcement officers and analysts have to explain issues and results to decision-makers and the public



DATA VISUALIZATIONS SHOULD...

- Graphically depict some aspect of the data
 - Either summarizing one variable, or showing how values of one variable (the dependent variable) vary by levels of another variable (the independent variable)
- Encourage the viewer to compare across categories/values of a variable



THE GRAPHICAL DISPLAY OF DATA

- Pie Charts
- Bar Charts/Bar Graphs
- Histograms



TYPES OF CHARTS

PIE CHART

A chart showing the differences in percentages among categories of a **nominal** (or **ordinal**) variable

- Categories are displayed as segments of a circle whose pieces add up to 100% of the total

BAR CHART

A graph showing the differences in frequencies or percentages among categories of a **nominal** or **ordinal** variable

- Categories are displayed as rectangles of equal width with their height proportional to the frequency or percentage of the category
- The taller the bar, the greater the frequency or percentage

HISTOGRAM

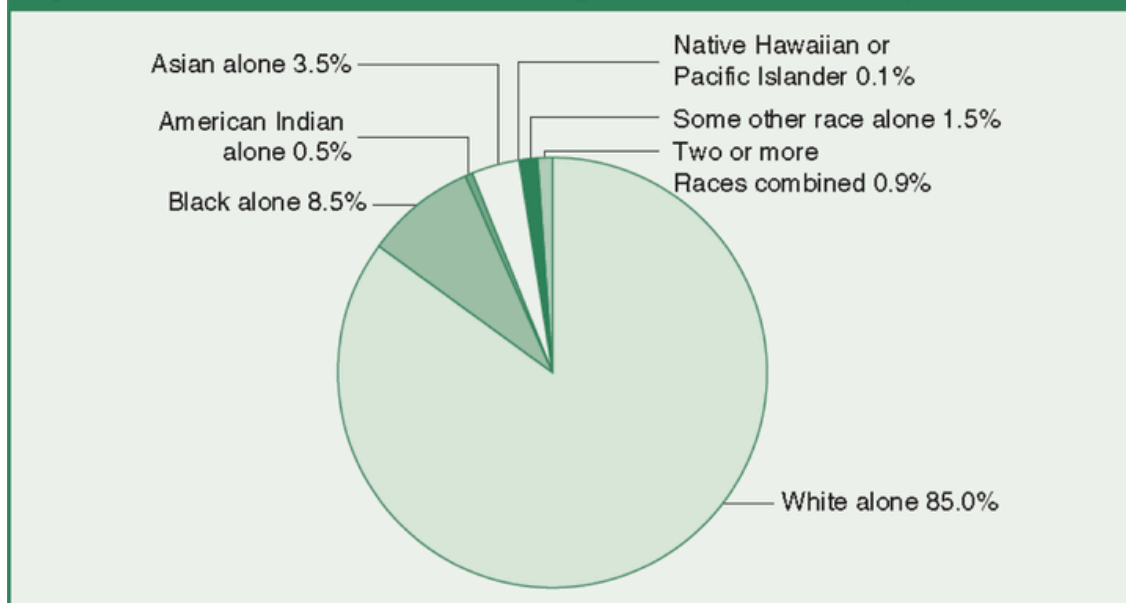
A special bar chart showing the differences in frequencies or percentages among categories of an **interval-ratio** variable

- Categories displayed as contiguous bars, with width proportional to the width of the category and height proportional to the frequency or percentage of that category



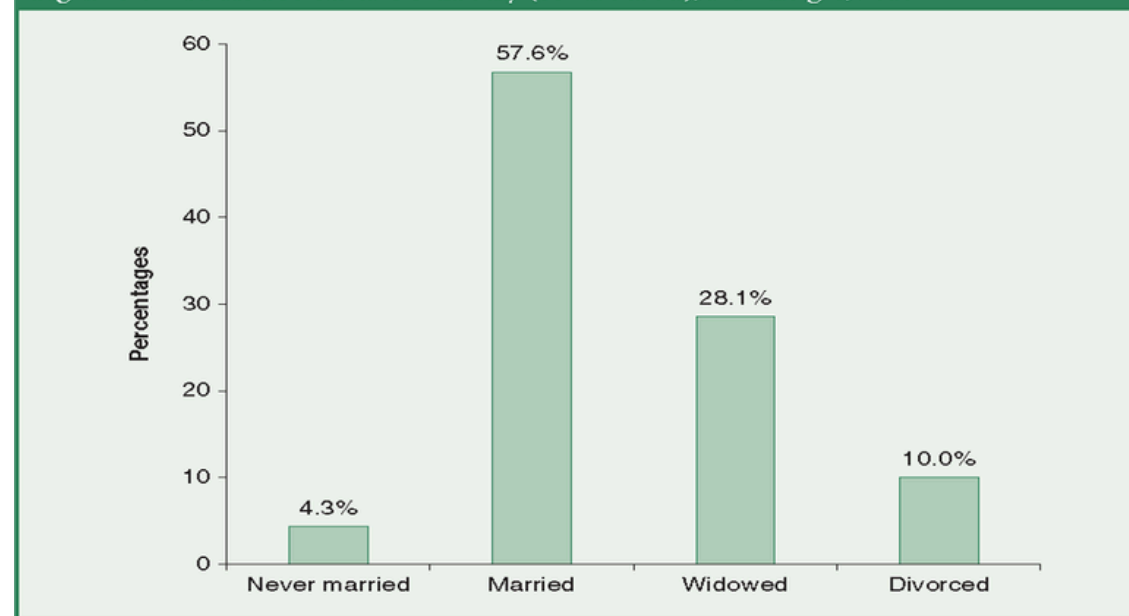
TYPES OF CHARTS

Figure 3.1 Three-Year Estimates of the U.S. Population 65 Years and Over by Race, 2009–2011



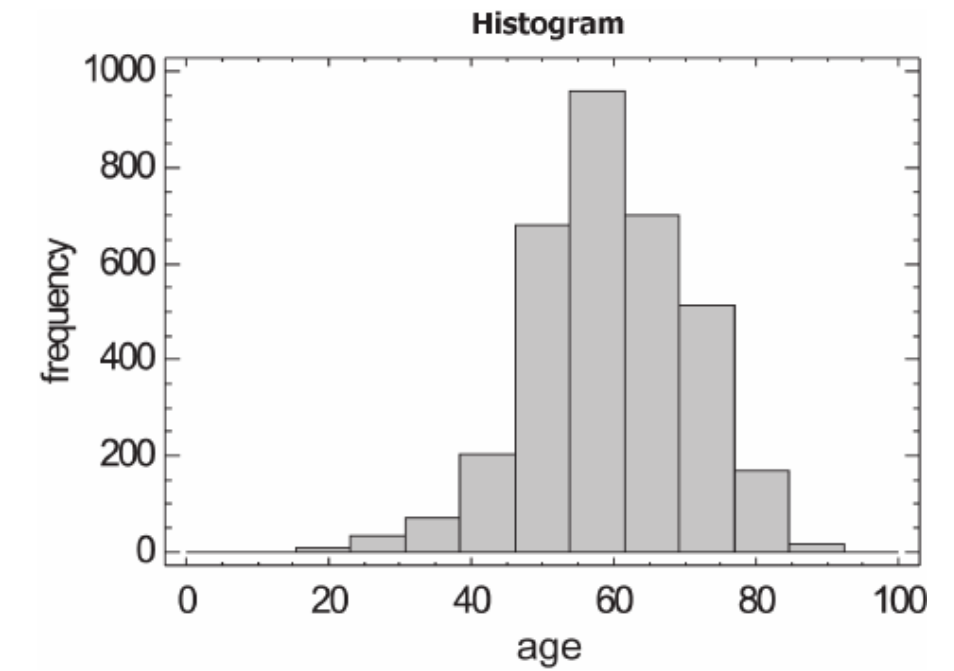
PIE CHART

Figure 3.3 Marital Status of U.S. Elderly (65 and older), Percentages, 2010



Source: U.S. Census Bureau, 2012 Statistical Abstract of the United States, Table 34.

BAR CHART



HISTOGRAM

LEVEL OF MEASUREMENT GUIDES VISUALIZATION

UNIVARIATE: CATEGORICAL

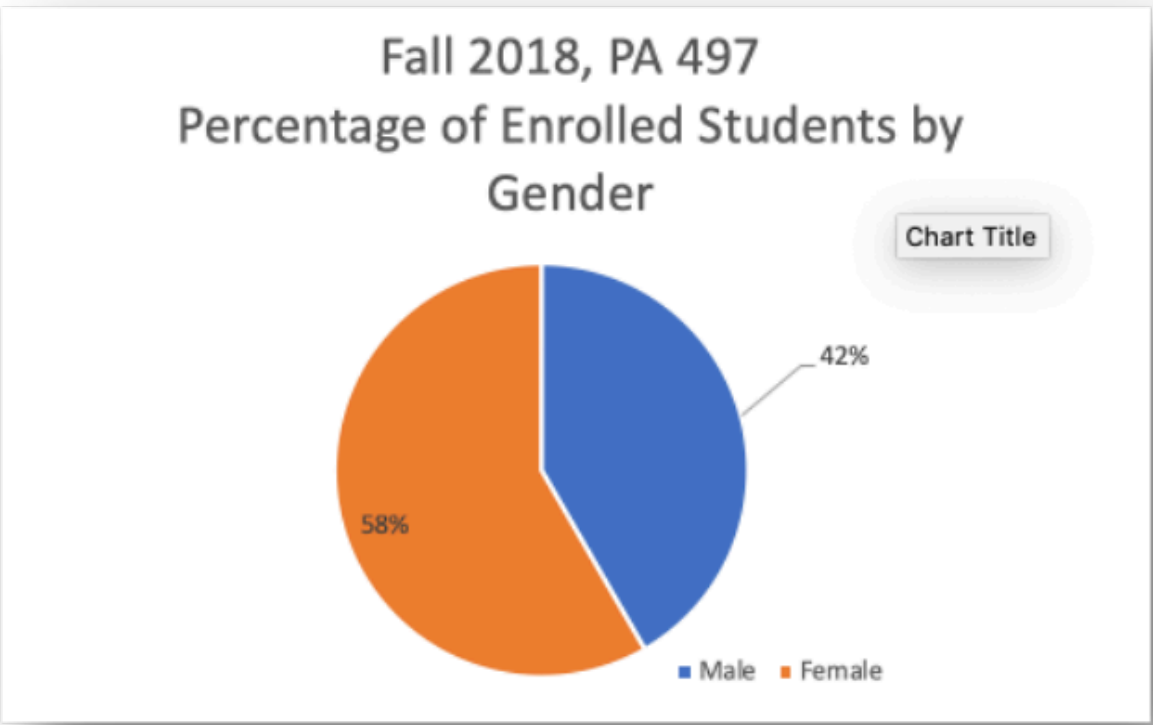
Table of Enrolled Students

Sex	Total f (%)
Male	5 (41.7%)
Female	7 (58.3%)
Total	12 (100%)

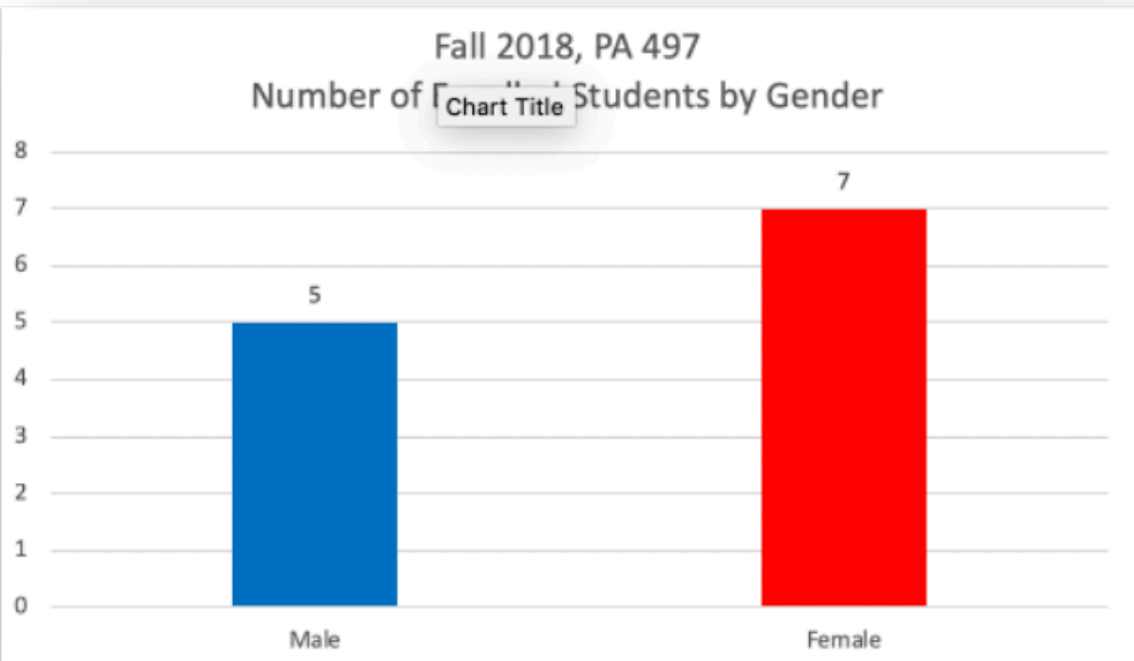
Sex is **categorical**,
specifically, nominal

Pie charts and bar charts
are best equipped for
categorical variables

Visualization #1
Pie Chart of Percent of Genders Enrolled



Visualization #2
Bar Chart of Frequencies of Genders Enrolled



LEVEL OF MEASUREMENT GUIDES VISUALIZATION

UNIVARIATE: CONTINUOUS

Table of Number of Nights in Previous Week Student Got 8 Hours of Sleep

# Days	Total <i>f</i>
1	14
2	16
3	10
4	5
5	2
6	2
7	1
Total	50

Days is **continuous**,
specifically, interval

Histograms are best
equipped for continuous
variables

Visualization #1
Histogram of Number of Nights in Previous Week
Student Got 8 Hours of Sleep



LEVEL OF MEASUREMENT GUIDES VISUALIZATION

BIVARIATE: CATEGORICAL DV, CATEGORICAL IV

Table of Enrolled Students' Major by Sex

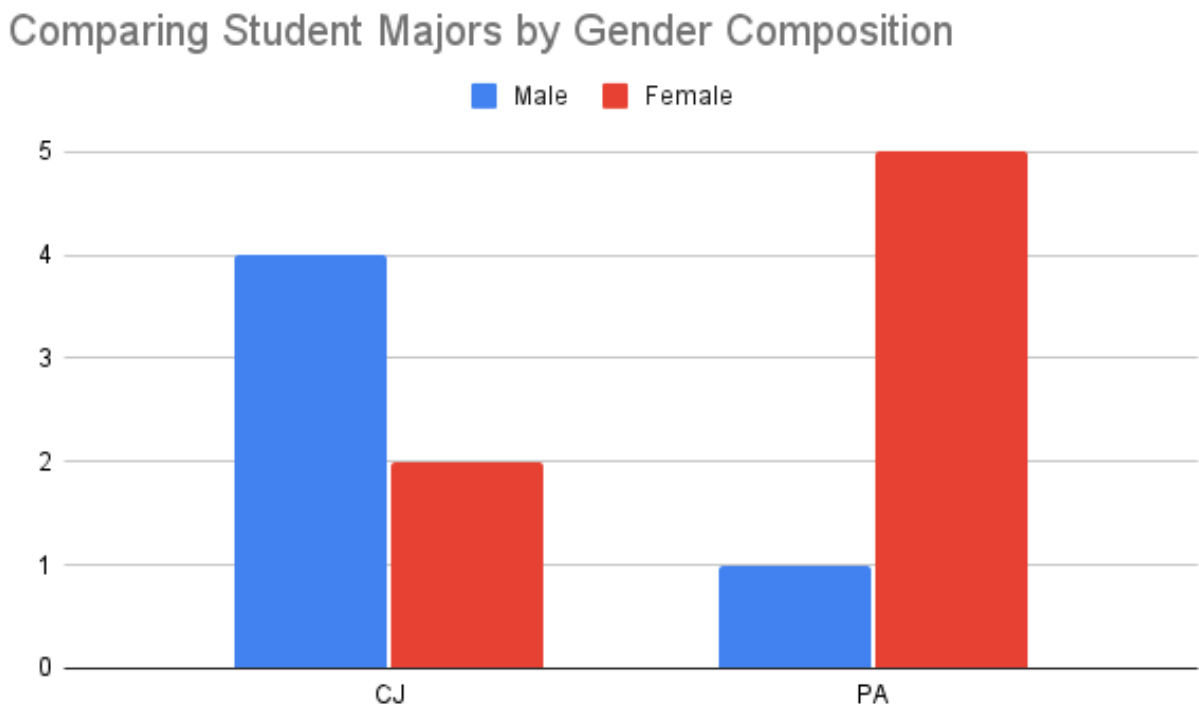
Sex	CJ f (%)	PA f (%)	Total f (%)
Male	4 (66.7%)	1 (16.7%)	5 (41.7%)
Female	2 (33.3%)	5 (58.3%)	7 (58.3%)
Total	6 (100%)	6 (100%)	12 (100%)

Sex is **categorical**, specifically, nominal

Major is also **categorical**, specifically,
nominal

Bar charts are best equipped for
comparing the values of two
categorical variables

Visualization #1
Bar Chart of Frequencies of Enrolled Students'
Major by Sex



LEVEL OF MEASUREMENT GUIDES VISUALIZATION

BIVARIATE: CONTINUOUS DV, CATEGORICAL IV

Table of Enrolled Students' Average Age by Sex

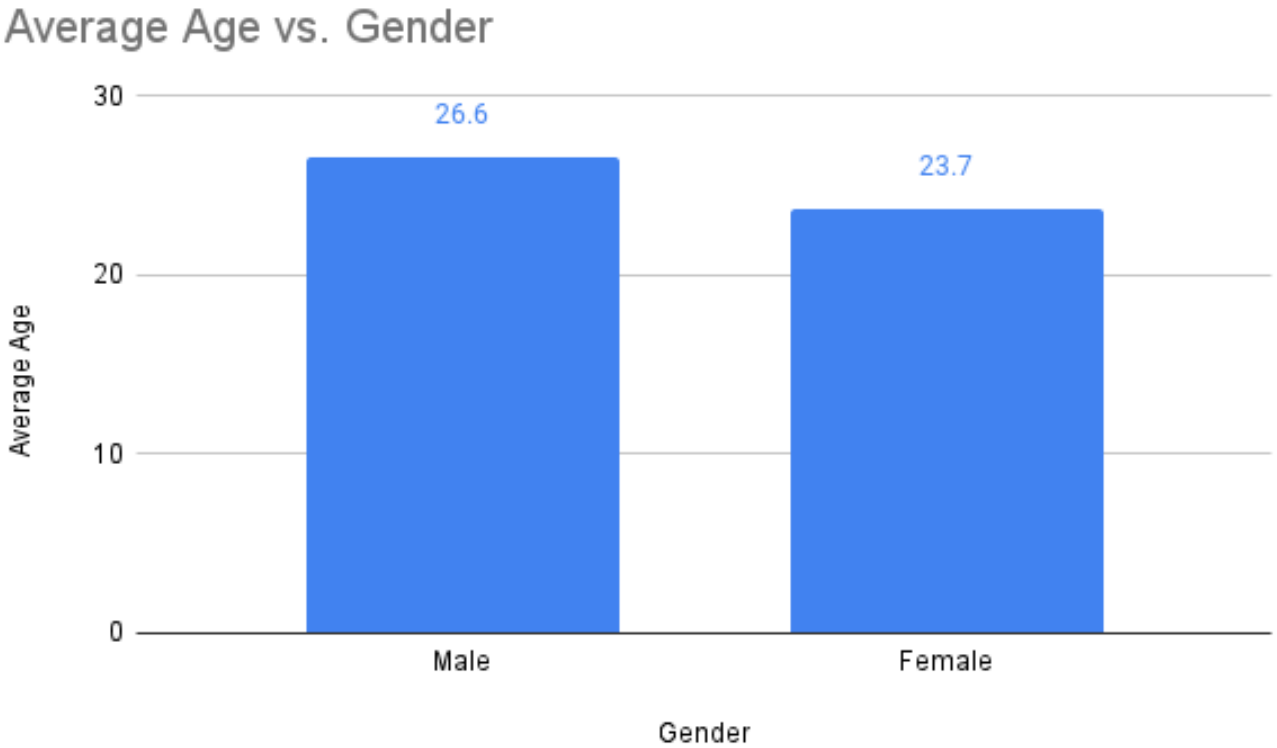
Sex	Average Age
Male	26.6
Female	23.7

Sex is **categorical**, specifically, nominal

Average Age is **continuous**, specifically, interval-ratio

Bar charts are best equipped for comparing the values of a continuous variable broken out by values of a categorical variable

Visualization #1
Bar Chart of Frequencies of Enrolled Students' Average Age by Sex



LEVEL OF MEASUREMENT GUIDES VISUALIZATION

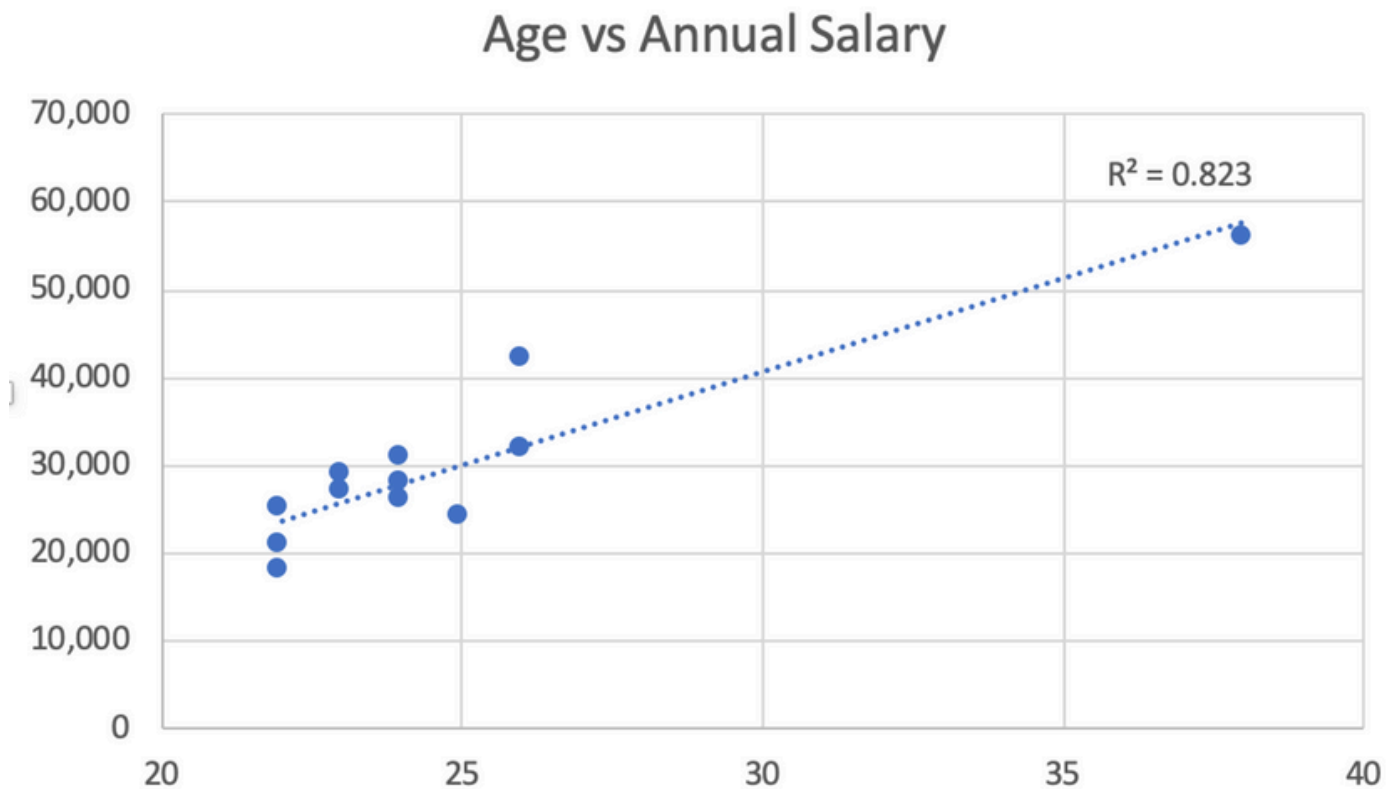
BIVARIATE: CONTINUOUS DV, CONTINUOUS IV

Age is **continuous**, specifically,
interval-ratio

Annual Salary is also **continuous**,
specifically, interval-ratio

Scatterplots are best equipped for
comparing the values of two
continuous variables

Visualization #1
Scatterplot of Salary by Age



INTERPRETING CHARTS

- Create charts in Google Sheets or Excel and copy-and-paste into Google Docs or Word
- Interpret charts in text, ideally next to the chart, so the reader can see what you are talking about
- If it's not obvious what charts are being referenced, you should mention it in the text:
E.g. "Table 3 shows that..."

CRIME TRENDS

In 2014, Sacramento experienced an eight-year low for Part I-related crime. In 2015, the city experienced a slight uptick from 2014 in both property and violent crimes. Last year, a reduction was seen again in both types of crimes. Excluding 2014, the 2016 property crime numbers are the lowest they have been in the last decade. The violent crime rate in 2016 is lower than the 10-year average.

